

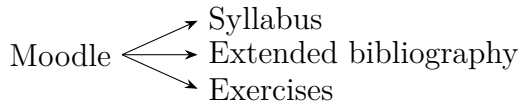
Andreas Goudelis

Indicative bibliography

- M. Blau “Lecture notes on General Relativity”
→ mostly chapters: 1–3, 8, 19, 23, 29, 33, 34
- B. Schutz “A first course in GR”
- Hobson, Efstathiou, Lasenby: “GR: An introduction for physicists”

Exam

→ No electronic devices

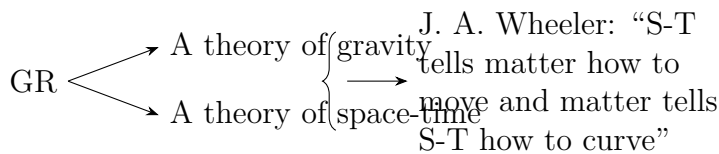
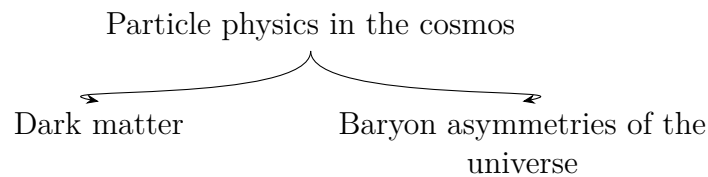


1) Introduction

- 4 fundamental interactions

The SM
 $SU(2)_L \times U(1)_Y$
electroweak theory

- i) EM: Maxwell’s eqs / {QED}
 - ii) Weak interaction:
 - iii) Strong: QCD
 - iv) Gravity: Newtonian / General relativity
- Why GR?
 - i) it is the 4th interaction
 - ii) It is not quantum yet
 - iii) Particle physics in the cosmos → shortcomings of the SM



2) Useful notions

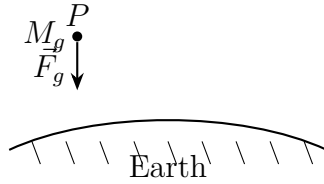
- Event: A physical occurrence in space and time (associated to a specific S-T point)
Remark: An event has no spatial extension and no duration.
Ex: Emission of a photon
– The presence of a point-like particle at some S-T point.
- Observers: Essentially, a reference frame
Note: we can switch reference frames through coordinate transformations
- Local time: Each observer has a notion of time and all observers measure time in the same way

- Worldline: The path that a point-like object follows in Space and Time

Note: For extended objects: world sheets and world volumes

3) The equivalence principle (EP)

A) Newtonian / Galilean version



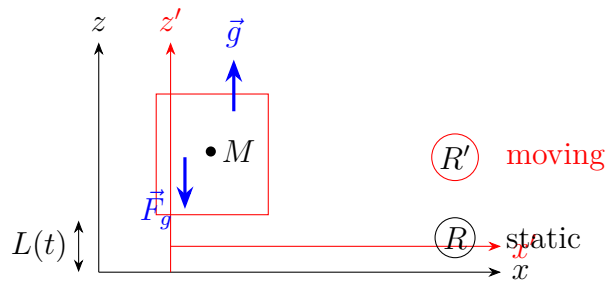
- From Newtonian gravity $\longrightarrow \vec{F} = -M_g \vec{\nabla} \phi, \quad M_g : \text{“gravitational mass”}$
- From Newtonian mech $\longrightarrow \vec{F} = M_i \vec{a}, \quad M_i : \text{“inertial mass”}$

$M_i \stackrel{?}{=} M_g$
yes, because we observe so

Galilean equivalence principle: $M_i = M_g$

$$\implies \vec{a} = -\vec{\nabla} \phi \quad \iff \quad \text{universality of free fall}$$

B) Einstein's version of the E.P.



- Case 0: Elevator in outer space, No forces $\implies P$ just floats
- Case 1: Still in outer space, elevator accelerated, w/ const ace $\vec{g}$
 $\implies P$ approaches the floor
- Case 2: Elevator brought in const. gravitational field
 $\implies P$ approaches the floor

$\longrightarrow$ Both R and R' agree on these statements

But R' cannot distinguish 1 from 2.

Gravity = Acceleration

$$L(t) = \frac{1}{2}gt^2 \quad \implies \quad \ddot{L}(t) = g \quad \text{and} \quad z' = z - \frac{1}{2}gt^2$$

$$t' = t \quad (\text{ignoring SR})$$

- In R : $\vec{F} = M \ddot{\vec{z}}$
 - In R' : $\vec{F}' = M \ddot{\vec{z}}' = M \ddot{\vec{z}} - M\vec{g} \implies \vec{F}' = \vec{F} - M\vec{g}$
- $\implies$ the acceleration of R' wrt R looks exactly like gravity in R' .

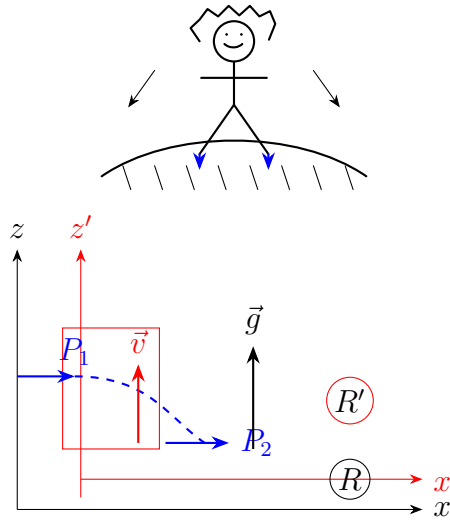
- Case 3: Still in the gravitational field and we cut the elevator rope

$\implies P$ floats for R' .

$\implies R'$ cannot tell if he/she is state in outer space or freely falling.

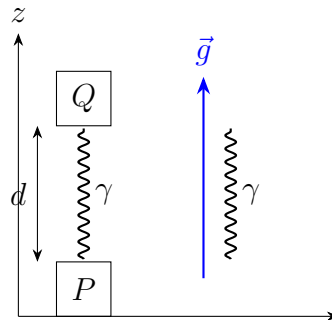
The equivalence principle: No physics experiment can distinguish locally between a uniform grav. field acting on the local inertial observer and a uniform acc. wrt that local inertial observer.

$\longrightarrow$ The one for whom SR holds



Exercise

- Gravitational redshift



P : emitter Q : Receiver

– photon of frequency f_P emitted from P

Q : frequency measured by Q (f_Q)?

S_Q/c^2 : Photon ToF $\Delta t = d/c$

During the same time the velocity of PQ increases by $\Delta v = g\Delta t = \frac{gd}{c}$.

In the acc. R.F. Q will see a doppler-shifted frequency f_Q

$$\frac{f_Q}{f_P} = 1 - \frac{\Delta v}{c} = 1 - \frac{gd}{c^2}$$

Equivalence Principle : Acceleration $\iff$ gravity

$\implies$ In the presence of a gravitational field, and in a fixed RF, photons will be redshifted as

$$\frac{f_Q}{f_P} = 1 - \frac{gd}{c^2}$$

Equivalently: clocks at Q tick slower than clocks at P .